

Pseudocode With Input, Process, And Output

Generated Jul 8, 2026 4:10 AM

What We'll Learn

By the end of this lesson, students can write pseudocode that clearly shows input, process, and output.

Words We'll Use

- Input
- Process
- Output
- Pseudocode
- Variable
- Compute
- Display

Before We Start

When solving a programming problem, how do we know what information is needed, what action must be done, and what result should be shown?

Let's Understand

Most simple programs follow the Input, Process, Output pattern.

Input means the data needed by the program.

Process means the action, formula, or computation done using the input.

Output means the result shown by the program.

Before writing pseudocode, it is helpful to identify these three parts first:

Pseudocode:

Input: What data is needed?

Process: What will the program do with the data?

Output: What result should be shown?

Basic pseudocode structure:

Pseudocode

START

INPUT value

COMPUTE result = process using value

DISPLAY result

END

A good pseudocode should show the correct order:

START → INPUT → PROCESS → OUTPUT → END

Let's Look at Examples

Example 1: Add Two Numbers

This example shows how to identify input, process, and output before writing pseudocode.

Problem:

Create pseudocode that asks for two numbers, computes their sum, and displays the result.

Problem Analysis:

Input: firstNumber, secondNumber Process: $\text{sum} = \text{firstNumber} + \text{secondNumber}$ Output: Display sum

Pseudocode

START

INPUT firstNumber

INPUT secondNumber

COMPUTE $\text{sum} = \text{firstNumber} + \text{secondNumber}$

DISPLAY sum

END

Explanation:

The program receives two numbers, adds them, stores the answer in sum, and displays the result.

Common mistake

Students may display sum before computing it.

Mini-check

What are the two inputs in this problem?

Example 2: Compute Quiz Percentage

This example shows how to use input values in a formula.

Problem:

Create pseudocode that asks for a quiz score and total number of items, then computes and displays the percentage.

Problem Analysis:

Input: score, totalItems Process: $\text{percentage} = \text{score} / \text{totalItems} * 100$ Output: Display percentage

```
Pseudocode
START
  INPUT score
  INPUT totalItems
  COMPUTE percentage = score / totalItems * 100
  DISPLAY percentage
END
```

Explanation:

The program gets the student's score and total items, computes the percentage, and displays it.

Common mistake

Students may forget to multiply by 100.

Mini-check

Why do we multiply by 100?

Example 3: Compute Total Cost

This example shows how input, process, and output work in a shopping-related problem.

Problem:

Create pseudocode that asks for the price of an item and the quantity bought, then computes and displays the total cost.

Problem Analysis:

Input: price, quantity Process: $\text{totalCost} = \text{price} * \text{quantity}$ Output: Display totalCost

```
Pseudocode
START
  INPUT price
  INPUT quantity
  COMPUTE totalCost = price * quantity
  DISPLAY totalCost
END
```

Explanation:

The program receives the item price and quantity, multiplies them, and displays the total cost.

Common mistake

Students may add the price and quantity instead of multiplying them.

Mini-check

What operation is used to compute the total cost?

Let's Practice

Identify the input, process, and output for each problem:

1. Ask for two numbers and display their sum.
2. Ask for a student's quiz score and total items, then compute the percentage.
3. Ask for the price and quantity of an item, then compute the total cost.
4. Ask for the length and width of a rectangle, then compute the area.
5. Ask for the user's first name and last name, then display the full name.

Now We Apply

Create pseudocode for a program that asks for a student's name, first quarter grade, and second quarter grade, then computes and displays the average grade.

Challenge Task

Create pseudocode for a program that asks for the number of hours worked and rate per hour, then computes and displays the total salary.

How Our Work Will Be Checked

- Correct identification of input
- Correct process or formula
- Correct output
- Proper order of steps
- Clear and readable pseudocode

Let's Reflect

Why should we identify the input, process, and output before writing pseudocode?

How We Show Learning

Students will submit one programming problem analysis and its complete pseudocode using the Input, Process, Output pattern.

Student Activities

Identify Pseudocode. Algorithm to Pseudocode. Complete Missing Lines. Trace with Values. Pseudocode Fix-It Task. Correction Log. Write pseudocode for area of rectangle. Write pseudocode for grade status. Exchange and revise with a partner.

Resources / References

JDK, IDE, projector or screenshots, printed worksheets, notebook, sample code snippets, and teacher-created answer guides.